

CHANANEL ZAGURI

Hardware-Oriented Software Developer | Python, Rust, C++ | Pre-Silicon Simulation

CONTACT

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SKILLS

Systems & Architecture:

- HW/SW Co-Design
- Hardware Modeling
- Software Infrastructure
- Simulators & Tools
- Performance Benchmarking

Performance & Optimization:

- SIMD & Vectorization
- Compiler Optimization
- Complexity Reduction
- Multithreading
- Linux Environment

Domains:

- ASIC
- Research Implementation
- Applied Cryptography
- Blockchain

Programming Languages:

- Python
- Rust
- Modern C++

LANGUAGES

- English (Professional Working Proficiency)
- Hebrew (Native)



WORK EXPERIENCE

Software Developer 2024 - PRESENT
Chain Reaction Ltd.

- Design and implement bit-accurate simulators for cryptographic hardware accelerators, collaborating closely with design and verification teams to refine architectures from concept to silicon.
- Bridge the gap between advanced cryptographic theory and functional systems by implementing and optimizing complex cryptographic algorithms with a strong emphasis on hardware acceleration (using Python and C++).
- Leverage Rust's safety and performance features to translate complex cryptographic designs into robust, highly scalable, and verifiable software models.

Automation Engineer 2023 - 2024
Chain Reaction Ltd.

- Built automated CI pipelines for nightly regression testing and reporting.
- Designed and developed automated test infrastructure synced with Jira, eliminating manual reporting and speeding up delivery.
- Designed and developed a GUI for post-production system testing with parallel test execution. Took ownership from design and development to deployment across lab environments and ongoing maintenance.

Student System V&V Engineer 2022 - 2023
Chain Reaction Ltd.

- Validated a full-stack ASIC Bitcoin mining platform, covering embedded firmware, blockchain protocols, and Web APIs.
- Owned the complete testing lifecycle, designing STPs and executing automated suites to monitor hash rates, thermals, and PSU reliability.

Research Assistant to Professor Eric Fredj 2021 - 2022
JCT

- Researched, maintained, and optimized a Python-based software simulator to model and analyze ocean current dynamics.



EDUCATION

Software Engineering (B.Sc.) - JCT 2019 - 2023
GPA 95, Dean's Award

High School Studies - Horev Jerusalem 2009 - 2013
5 units of mathematics - 97
Majors - Physics, Computer Science, and Economics